

Resolution and MPI scaling for multiphase flows

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Abstract

We study the CPU scaling of Basilisk for the stock Poisson and Laplacian tests, Marangoni-driven migration, bubble bursting, elastic Taylor–Culick retraction and drop impact. The measurements on MareNostrum 5 and Snellius extend to 16,384 MPI ranks for the stock kernels. For Marangoni migration, we also compare the drop velocity with its analytical value and find that the error does not decrease monotonically at the finer resolutions. At fixed resolution, the wall time initially decreases with increasing rank count, then levels off and can increase again. The location of this knee changes with the mesh and the flow being calculated. In a research project, we first establish the resolution needed for the physical quantity of interest, then use a scaling test to choose ranks near the knee. Grid independence at one set of parameters cannot be assumed at another, nor can the rank count chosen for one mesh be assumed suitable after refinement. Both tests need to be repeated as the parameter study develops, particularly where smaller length scales or new flow regimes appear. The same applies to regularizations and cutoffs introduced when solving the continuum equations. A change in the numerical model or computing system also requires a new scaling test.

1 Grid sensitivity and parallel scaling

A grid-independence study must follow the flow far enough to test the quantity of interest. During bubble bursting, the cavity can collapse into a jet much thinner than the bubble radius. Similarly, a drop that is well resolved before impact can spread into a poorly resolved lamella. A mesh chosen from the initial geometry alone may miss these later stages. Moreover, convergence of the drop trajectory need not imply convergence of the peak impact force or of a jet radius. The required resolution depends on which of these quantities we calculate and how accurately we need it.

The grid is only one of the choices made when we solve the continuum Cauchy equations numerically. We also choose the time-step restriction, solver tolerances and domain size, and may regularize a constitutive law or introduce a cutoff. In a regularized viscoplastic model, the effective viscosity may be capped at a finite value. The nominally unyielded region then has a large but finite viscosity. Refinement at a fixed cap tests convergence for that regularized equation; the viscosity cap must be varied separately. The criterion used to identify flow arrest needs a similar check. We tested both in our study of bursting bubbles in a viscoplastic medium [1]. Even after these numerical tests, comparison with theory and experiments is needed to assess the physical predictions of the model.

These checks cannot be settled for an entire project by testing one case at the outset. A change in the physical parameters can produce a thinner jet, steeper gradients or a different flow regime, for which the same mesh is insufficient. Increasing the resolution also changes the amount of work assigned to each MPI rank. Once we have established an adequate resolution, we therefore measure the wall time at several rank counts. We seek the *knee*, where the curve bends from approximately

inverse scaling towards a plateau. Around this bend, doubling the rank count no longer halves the wall time. Its location must be checked again as the calculation changes.

2 Numerical method and timing

For each strong-scaling test, we vary the number of MPI ranks N_{MPI} while keeping the grid resolution, geometry, material parameters and output settings fixed. We use uniform grids with the Basilisk tree backend and disable mesh adaptation during these timings. The plots give wall time per iteration; $T \propto N_{\text{MPI}}^{-1}$ denotes ideal scaling. The separate Marangoni accuracy study uses mesh adaptation.

We use the stock Poisson and Laplacian tests without modification [2, 3], taking the mean time per kernel operation over each test’s timing loop, together with the time spent in MPI routines. For retraction and impact, the timer covers ten solver steps after geometric initialization. The bursting case starts from an interface constructed separately, and the ten-step timer begins after restoring it. Neither construction nor restoration enters this time. Checkpoint output is disabled for all three cases. When selecting ranks for a full simulation, we also need timings from the relevant stages of the flow with the output that the simulation will write.

For Marangoni migration, we divide the elapsed solver time by the number of steps in the prescribed interval. At 256 points per radius, the interval is $t/t_0 = 0\text{--}0.05$ for 2–256 ranks and 0–0.5 for 512–1,024 ranks. Thus the averages on either side of this change cover different flow histories.

Each configuration has been timed once. The curves identify where scaling improves or deteriorates, but neighbouring points on a nearly flat curve should be repeated before choosing between them. The code, timing tables and plotting scripts are available at <https://github.com/comphy-lab/hpc-basilisk-scaling>. To compare runs between systems, we record the source revision, compiler flags and MPI implementation, together with the node type, ranks per node and rank binding.

3 Poisson and Laplacian tests

In an incompressible-flow calculation, the velocity obtained after advection, viscous stresses and interfacial forces must be projected onto a divergence-free field. Basilisk uses a multigrid Poisson solver for this step. Restriction and prolongation connect the different grid levels, and MPI communication exchanges data between subdomains. At a fixed grid size, each rank has fewer cells as the number of ranks increases, while these exchanges are still required. The stock tests measure the resulting scaling before interface reconstruction or constitutive evolution is added.

For the two-dimensional test, we use a full quadtree at $L = 14$, containing 2^{28} cells. The Snellius Poisson time decreases up to 8,192 ranks and levels off at 16,384 ranks (Figure 1(a)). As the wall time approaches this plateau, communication takes a larger fraction of it. We include the published Curie data for comparison with the same test.

The three-dimensional test uses a full octree at $L = 9$, with 2^{27} cells, and is compared with the published Occigen data. On Snellius, the Poisson time decreases from 19.34 s at two ranks to 0.0390 s at 4,096 ranks. Increasing to 16,384 ranks reduces it to 0.0333 s (Figure 1(c)). Thus four times as many ranks save only about 15% of the wall time over this interval. The Laplacian takes less time per iteration on the same grid, and its curve has a different dependence on the rank count.

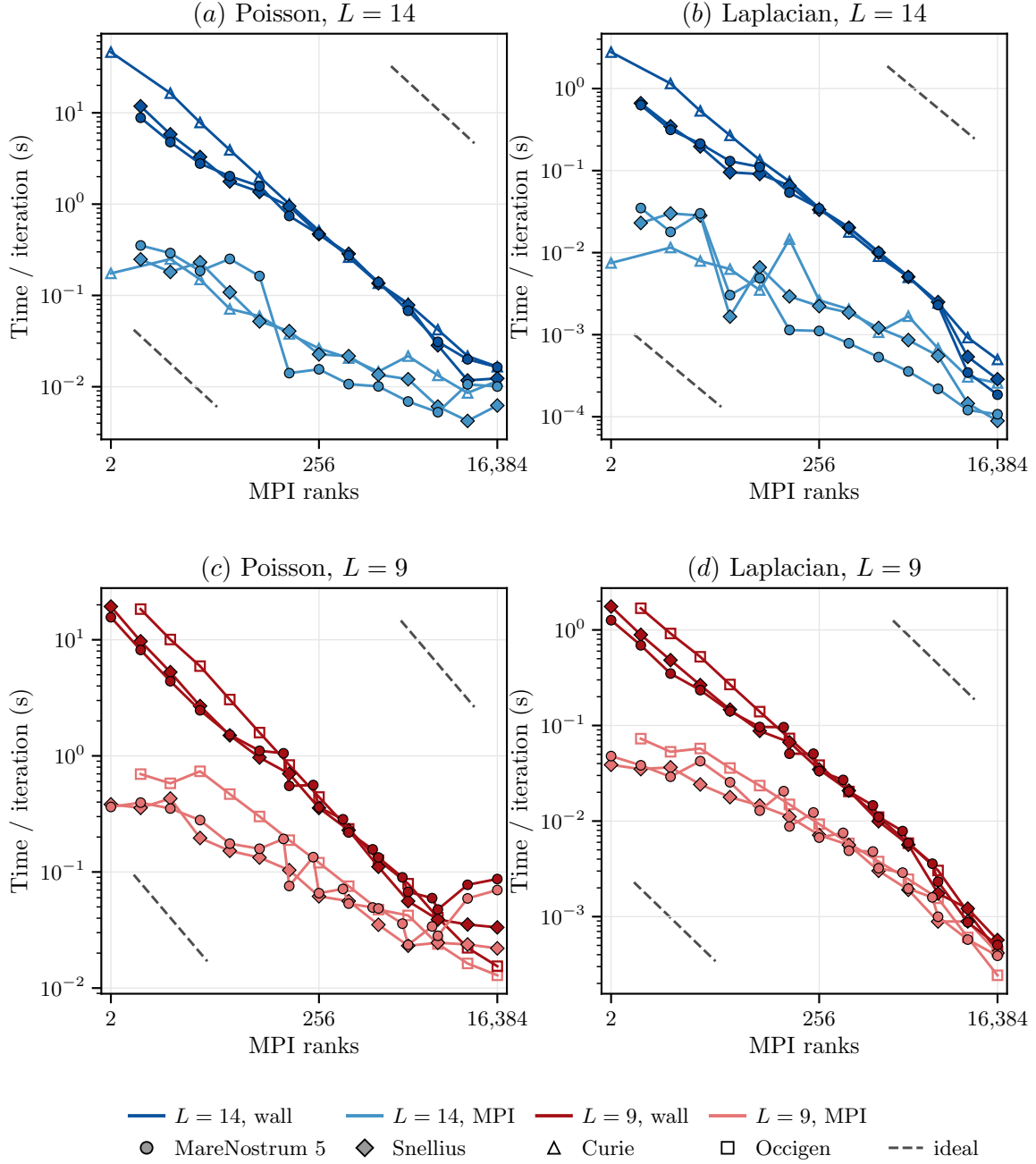


Figure 1: Time per iteration for the stock Poisson (left) and Laplacian (right) tests. (a,b) Full two-dimensional quadtree at $L = 14$. (c,d) Full three-dimensional octree at $L = 9$. The $L = 14$ results are shown in blue and the $L = 9$ results in red, with darker shades for wall time and lighter shades for MPI time. Circles denote MareNostrum 5 and diamonds Snellius. The open symbols are the published Curie (two-dimensional) and Occigen (three-dimensional) data. Solid lines join the measured points. The grey dashed segments at the lower left and upper right show $T \propto N_{\text{MPI}}^{-1}$; their vertical positions are arbitrary.

4 Marangoni migration

Here, the drop moves under a prescribed surface-tension gradient. We use the conservative, balanced formulation of Abu-Al-Saud, Popinet and Tchelepi [4]. The analytical migration velocity gives us a reference against which to test the calculated drop speed as the mesh is refined.

Figure 2 shows the timings for two series of calculations. In the axisymmetric case, we increase the resolution from 64 to 512 points per drop radius. In the planar case, we increase the number of drops from one to 32, keeping 64 points per radius. Both series extend to 1,024 ranks. The curves for the smaller calculations flatten while the larger ones still decrease. Increasing either the resolution or the number of drops changes how much computation can be distributed over the MPI ranks. A rank count near the plateau of a coarse single-drop case can still lie on the decreasing part of a finer-grid or multiple-drop curve.

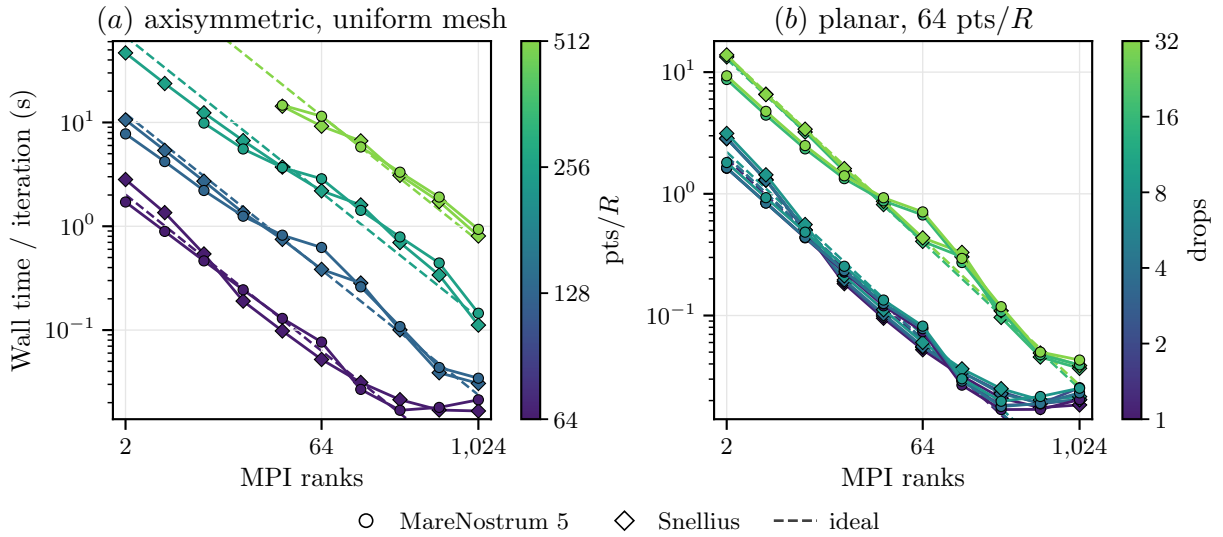


Figure 2: Wall time per solver step for Marangoni-driven migration on MareNostrum 5 GPP (circles) and Snellius Genoa (diamonds). (a) Axisymmetric calculations with 64–512 points per drop radius. (b) Planar calculations with 1–32 drops, each resolved by 64 points per radius. The dashed lines give the ideal scaling. For integral surface-tension formulations applied to Marangoni flow, see Ref. [5].

We compare the migration velocity with the Young–Goldstein–Block prediction and the Basilisk reference data in Figure 3. The relative error decreases rapidly between eight and 32 points per radius. Beyond 32 points per radius, it varies non-monotonically with refinement. The N^{-2} line is a guide for the coarse-grid points; fitting this slope through all the resolutions would obscure the change in behaviour at the finer grids. The drop-frame velocity fields show the flow developing during migration.

Although the larger grids scale over more ranks, they do not give a monotonic reduction in the velocity error. Domain size, integration time and solver tolerance are other possible sources of error and need separate tests. We cannot infer accuracy from the improvement in parallel scaling.

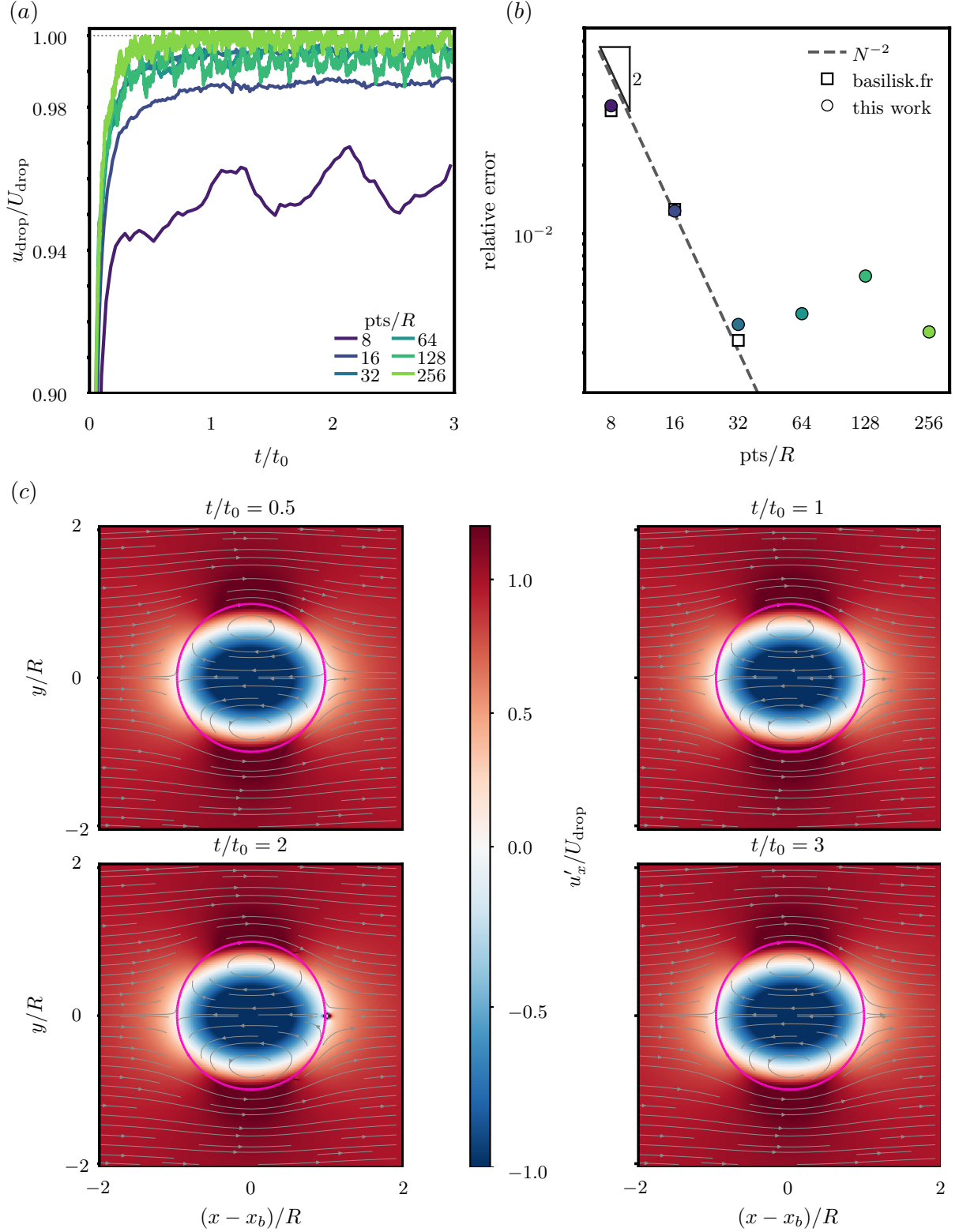


Figure 3: Marangoni-driven migration under a prescribed surface-tension gradient. (a) Speed relative to the Young–Goldstein–Block prediction at 8–256 points per radius. (b) Relative error and an N^{-2} guide. (c) Drop-frame axial velocity at 64 points per radius, shown at $t/t_0 = 0.5, 1, 2$ and 3 . Magenta marks the interface. See Ref. [5] for the interfacial-stress formulation.

5 Bursting, retraction and impact

We next consider bubble bursting, elastic Taylor–Culick retraction and Newtonian drop impact, using the same two-phase solver infrastructure. The three axisymmetric cases use $L = 9, 10$ and 11 , corresponding to 512^2 , 1024^2 and 2048^2 cells. The coarser grids are tested over 2–768 ranks. At $L = 11$, the tests start at eight ranks and reach 768 ranks for bursting and 2,304 ranks for retraction and impact.

At $L = 10$, the three cases have similar scaling ranges despite their different times per step (Table 1). On the coarser $L = 9$ grid, the time for bursting levels off at about 0.008 s beyond 96 ranks, while retraction and impact approach 0.010 s and 0.011 s, respectively. Refinement changes this range. The same rank count can consequently fall on a decreasing part of one curve and near the plateau of another, depending on both the mesh and the flow being calculated.

Kernel	Wall time per iteration (s), $L = 10$		
	2 ranks	192 ranks	768 ranks
Bursting bubble	1.28	0.0174	0.0108
Elastic Taylor–Culick	1.85	0.0231	0.0127
Newtonian drop impact	2.04	0.0267	0.0173

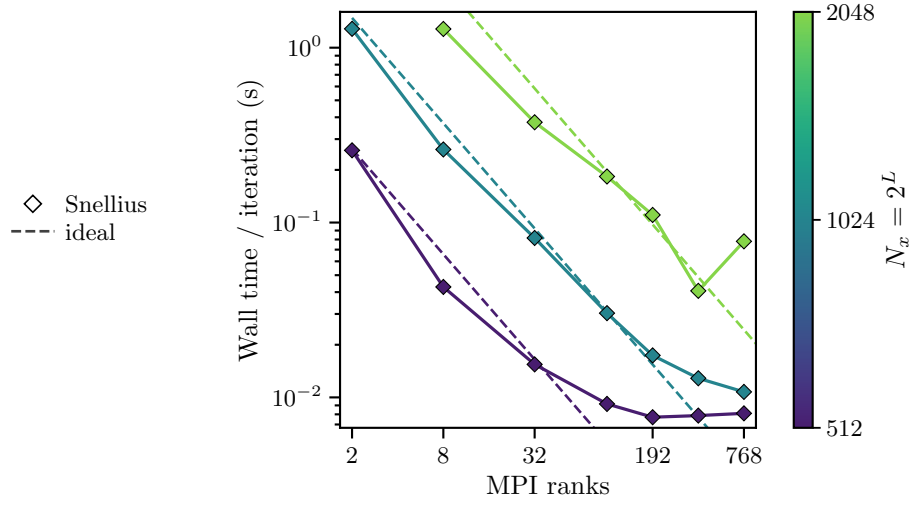
Table 1: Wall time per iteration on Snellius Genoa for the three axisymmetric cases.

For the finest bursting case, the time decreases to 0.0407 s per step at 384 ranks and then rises to 0.0781 s at 768 ranks (Figure 4). Doubling the number of ranks nearly doubles the time in this part of the curve. The elastic sheet includes log-conformation extra-stress evolution and continues to become faster over the larger rank range: the $L = 11$ time falls from 0.0341 s at 768 ranks to 0.0193 s at 1,536 ranks and 0.0152 s at 2,304 ranks (Figure 5). The final increase in ranks still reduces the time, although by a smaller amount than the preceding one.

For Newtonian impact, the drop is initially at $(1.05, 0)$ in a box of side 8, moving towards the left wall, with the axis of symmetry along the bottom boundary. At $L = 11$, we obtain 0.0339 s per step at 768 ranks, 0.0250 s at 1,536 ranks and 0.0423 s at 2,304 ranks (Figure 6). Hence, increasing the rank count from 1,536 to 2,304 helps the retraction calculation but slows the impact calculation. The largest rank count reached by a test is not necessarily a useful one for that flow. These application timings do not separate computation, communication and other overheads, so they do not determine which contribution causes the departure from ideal scaling.

The sequences in Figures 4–6 are from separate completed simulations. They show the narrowing cavity and emerging jet, the growing rim of the retracting sheet, and the thin lamella produced by drop impact. A resolution study must include these stages if the jet, rim or lamella determines the result of interest. Within each sequence, the spatial scale and colour normalization are fixed. The frames are ordered from left to right, then top to bottom.

(a)



(b)

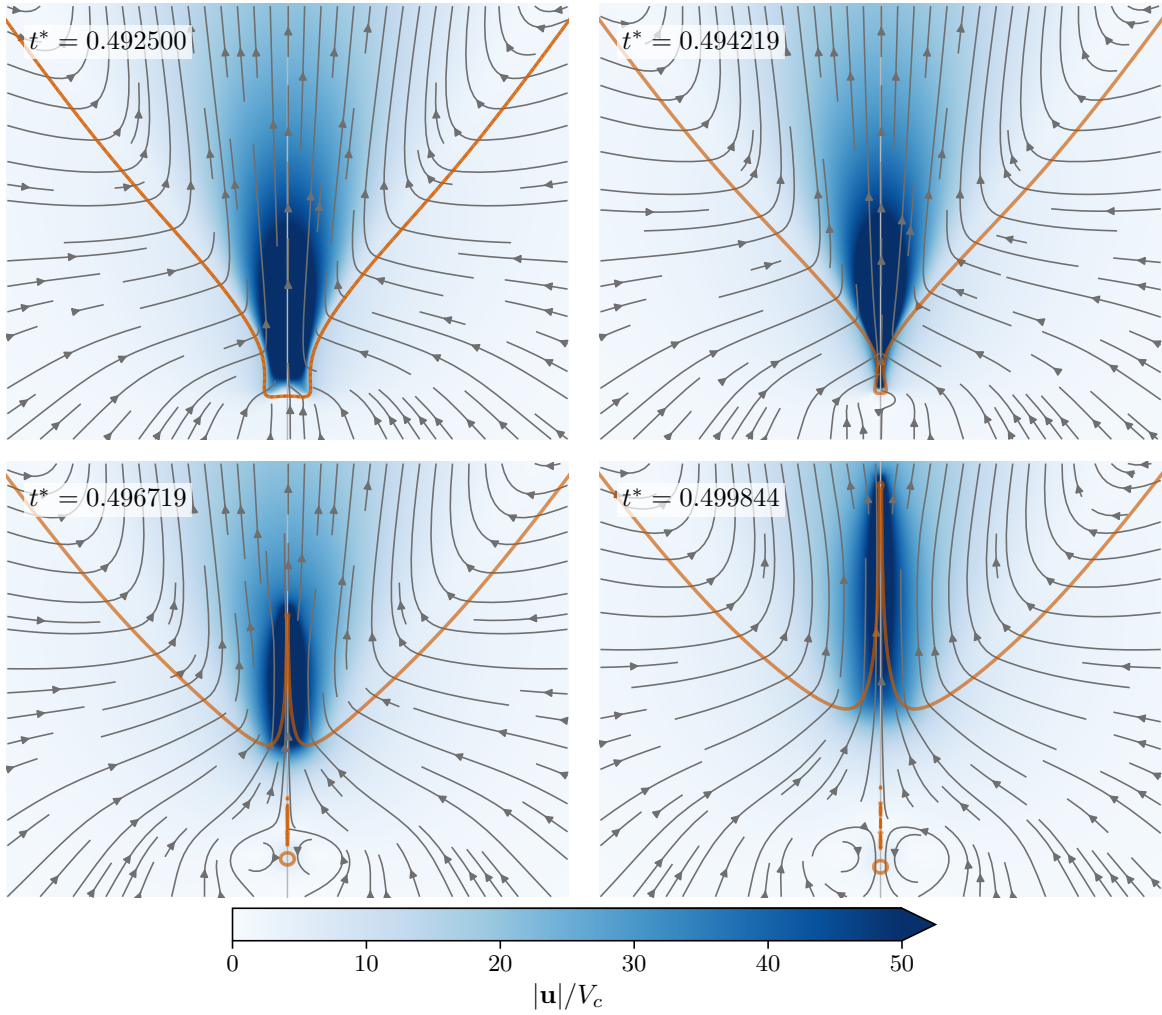
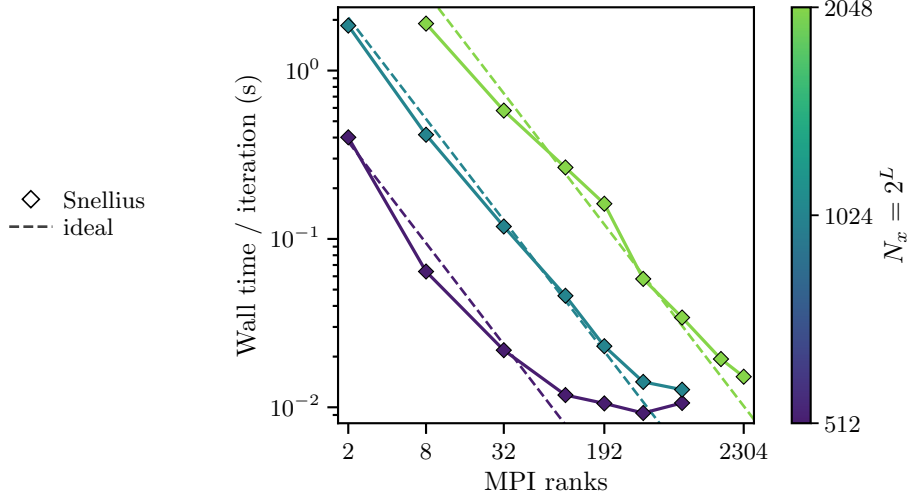


Figure 4: Bursting bubble [6, 1]. (a) Snellius Genoa strong scaling of the axisymmetric calculation; colour gives $N_x = 2^L$ and dashed lines show ideal scaling. (b) Jet inception [7] at $Oh = 0.03$, $Bo = 0$. Blue gives $|u|/V_c$, where $V_c = \sqrt{\sigma/(\rho_l R)}$ and $t^* = tV_c/R$; grey curves are instantaneous streamlines and orange marks the interface. The sequence shows the narrowing cavity, emerging jet and entrapped bubble. Colours saturate above $50V_c$.

(a)



(b)

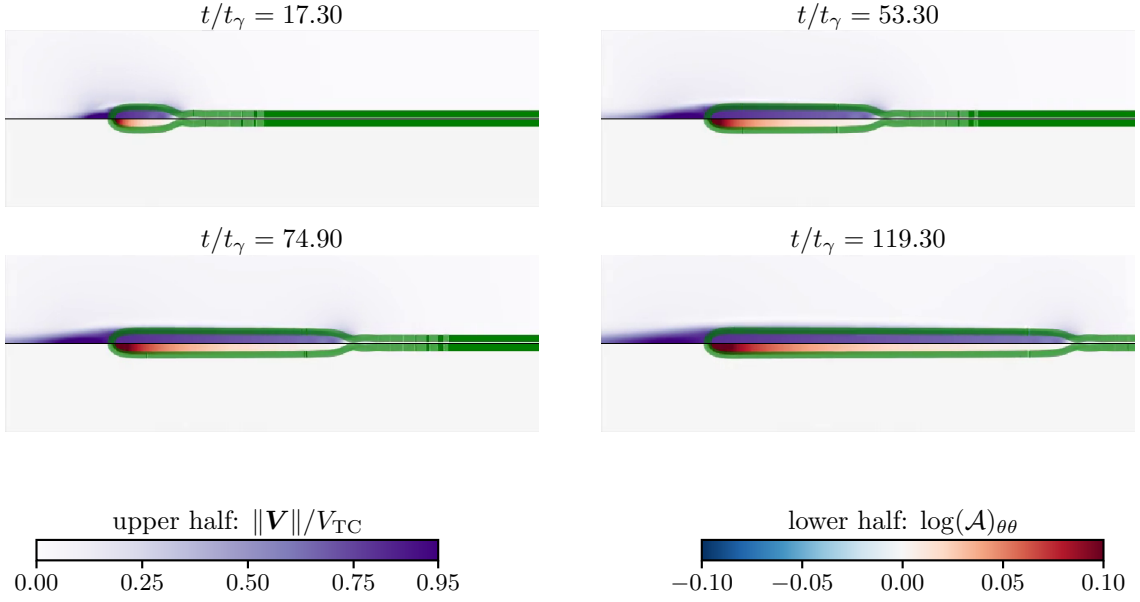
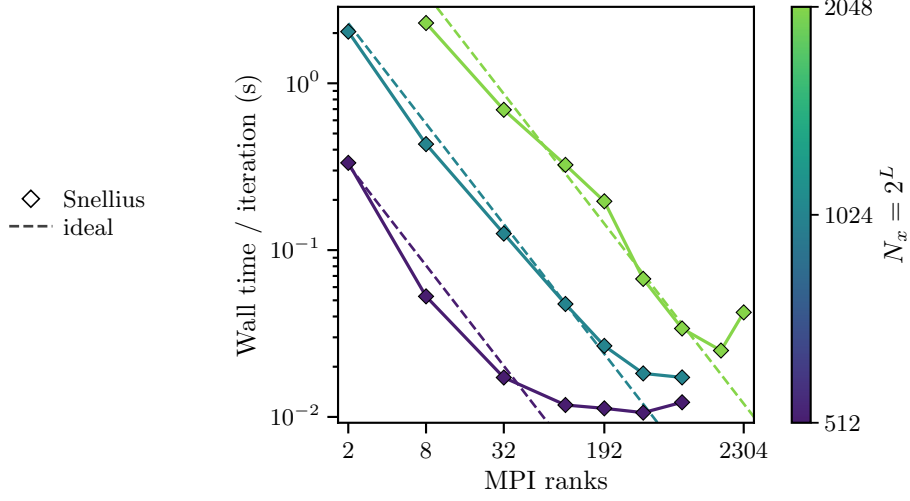


Figure 5: Elastic Taylor–Culick retraction. (a) Snellius Genoa strong scaling of the axisymmetric calculation. (b) Rim evolution at $Ec = 0.1$, $Oh = 0.05$ in a relaxation-free calculation. The view follows the rim. Each frame combines speed relative to the Taylor–Culick velocity (upper half) and the azimuthal log-conformation ($\log \mathcal{A})_{\theta\theta}$ (lower half); green marks the interface. Here $t_\gamma = \sqrt{\rho H_0^3 / \sigma}$, with H_0 the initial sheet thickness. For retraction and the influence of the surroundings, see Ref. [8].

(a)



(b)

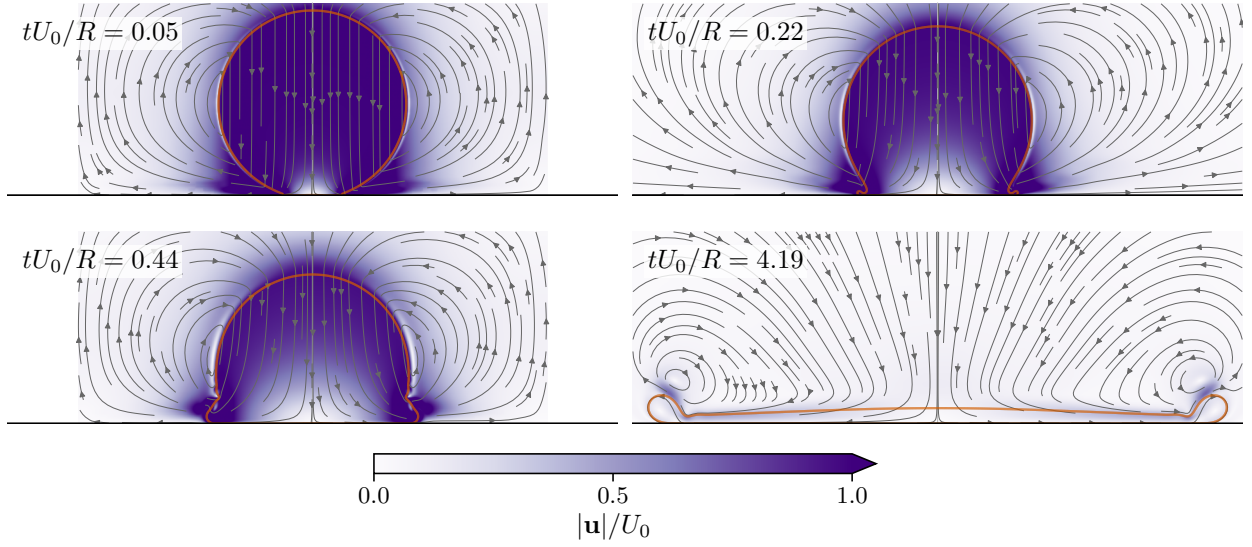


Figure 6: Newtonian drop impact. (a) Snellius Genoa strong scaling of the axisymmetric calculation. (b) Impact and spreading at diameter-based $We_D = 200$ and $Oh_D = 0.01$. Lengths are scaled by the initial drop radius R and time by R/U_0 , where U_0 is the impact speed. Purple shows speed, grey curves are instantaneous streamlines, orange marks the interface and black marks the substrate. Colours saturate above U_0 . All four frames retain the same view, including the extended lamella at $tU_0/R = 4.19$. Related studies address free-surface wave focusing [9] and drop-impact forces and spreading [10, 11].

6 Choosing resolution and MPI ranks

We prefer to run near the knee because the time saved by further increasing the number of ranks becomes small. The shortest wall time can still be useful for an urgent calculation, but for a parameter study those additional cores may be better used to run another case. We quantify the resource use by the allocated core time. If C_{alloc} cores are reserved for a wall time t_{wall} , then

$$\mathcal{C} = C_{\text{alloc}} t_{\text{wall}}. \quad (1)$$

Here C_{alloc} counts all reserved cores, including any that remain idle. It equals N_{MPI} when each allocated core runs one rank. If the scheduler reserves whole nodes, leaving some of their cores idle does not reduce the allocation. We therefore compare both wall time and allocated core time when selecting ranks near the knee.

We start with a grid study for the physical quantity and accuracy we need. This includes mesh and time resolution, together with the relevant solver tolerances, regularizations and cutoffs, over the stages of the flow that set that quantity. At an adequate resolution, a short MPI test should extend far enough to include the decreasing and flattening parts of the time curve. We then repeat points near the apparent knee, check the memory requirement, and choose a rank range from the measured wall and core times. Before using it for a production run, the timing test must cover representative flow stages with checkpointing and the other required output.

As the parameter study develops, we repeat these checks where the flow introduces new length or time scales, where the model changes, or when we move to another computing system. The sensitivity results should remain associated with the cases for which they were obtained. In particular, near a transition we may need to establish whether a jet forms or a drop detaches. Agreement in a global error measure is insufficient if the calculations disagree on that outcome. Tests near such transitions can be more informative than another check of the case used at the start of the project. If the resolution required for the result is too expensive, we must reconsider the numerical method, the model or the extent of the study.

The present kernel tests reach 16,384 ranks, equivalent to about 85 fully populated nodes with 192 cores each. The Marangoni cases reach 1,024 ranks, while retraction and impact reach 2,304. A comparison at 64 complete nodes, or above 100 nodes, requires measurements at the corresponding rank counts and a sufficiently large problem. For a machine benchmark, it is reasonable to increase the problem size to measure performance at those scales. For a research calculation, the additional resolution or larger domain must also be required by the physical question.

7 Conclusion

For the three-dimensional Poisson test, increasing from 4,096 to 16,384 ranks saves only about 15% of the wall time. Among the finest axisymmetric cases, retraction continues to become faster at 2,304 ranks, while impact has already slowed down and bursting turns upwards much earlier. The larger Marangoni calculations also scale over more ranks, but the velocity error does not decrease monotonically with refinement. Thus the resolution needed for an accurate result and the ranks needed to compute it efficiently have to be tested separately. We choose the numerical parameters from the accuracy of the physical quantity, then select ranks near the scaling knee for that calculation. As the study reaches new regimes, both tests need to be repeated. They belong within the parameter study, rather than being treated as checks completed once at the start of the project.

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